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Autonomous Vehicle: Application of Low, Mid, and High Level Computer Vision for Road Navigation

ITA-0513-Computer Vision

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Autonomous Vehicle & Visual Perception

What Enables a Vehicle to “See”?

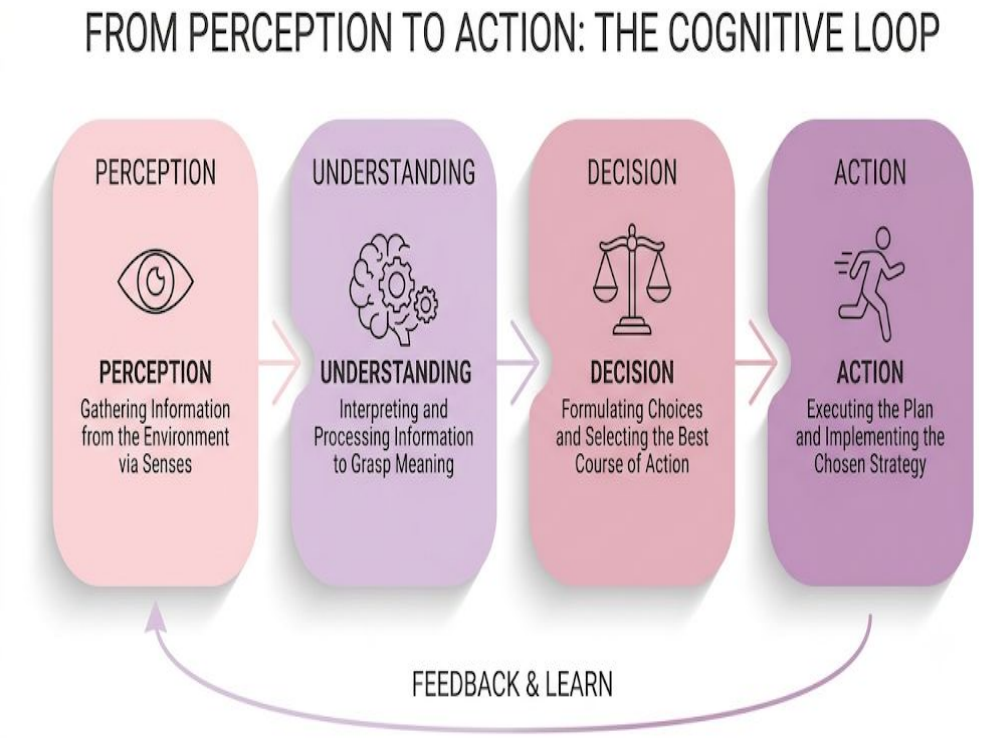
An autonomous vehicle must continuously perceive its environment before making a driving decision.

Primary visual inputs:

- Road and lane markings
- Vehicles and pedestrians
- Traffic signals and signs
- Road boundaries and obstacles
- Environmental conditions

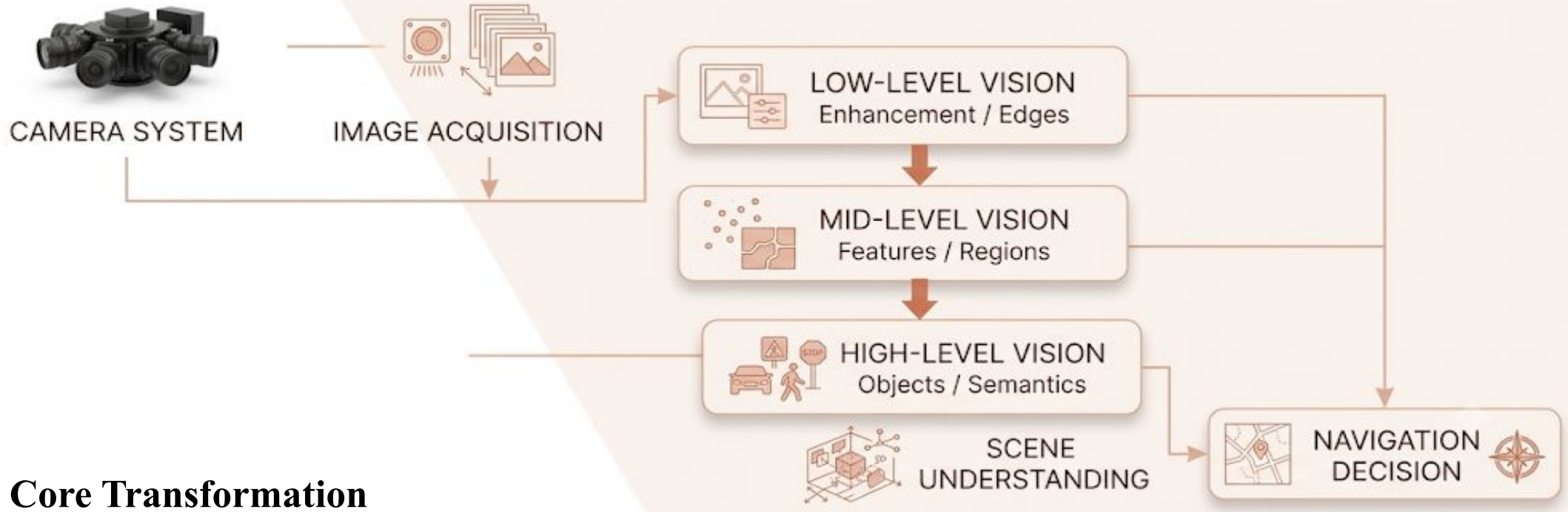
Computer Vision Role:

Transforms camera-captured visual data into structured information for autonomous decision-making.



Computer Vision Architecture :

End-to-End Visual Perception Pipeline



Core Transformation

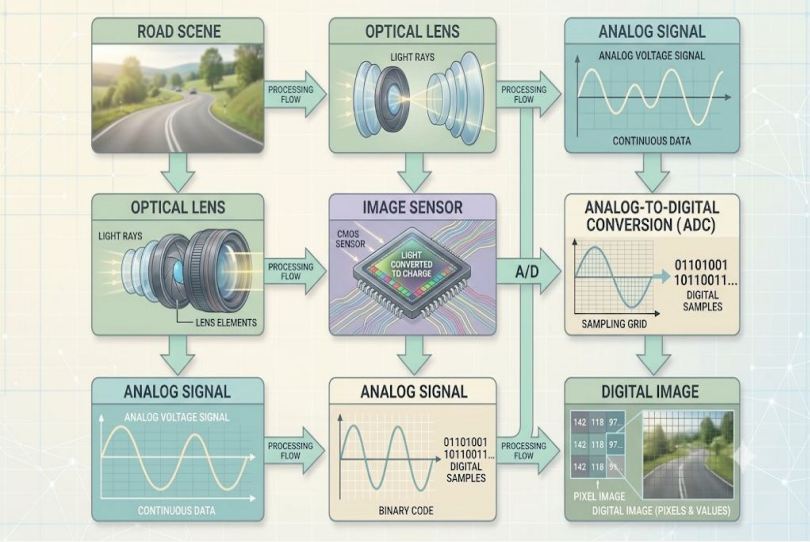
Pixels → Features → Objects → Scene → Decision

Image Acquisition & Camera Input :

Stage 1: Capturing the Driving Environment

The camera system converts the physical environment into digital image frames.

Acquisition process:



Important Parameters

- Resolution
- Frame rate
- Field of view
- Exposure
- Dynamic range
- Image format

Example:

A forward-facing camera continuously captures road frames at a suitable frame rate for real-time perception.

Low-Level Image Processing :

Stage 2: Low-Level Image Processing

Low-level vision operates directly on **pixel-level information**.

Major Operations

1. Noise Reduction

Removes unwanted variations from captured images.

2. Image Enhancement

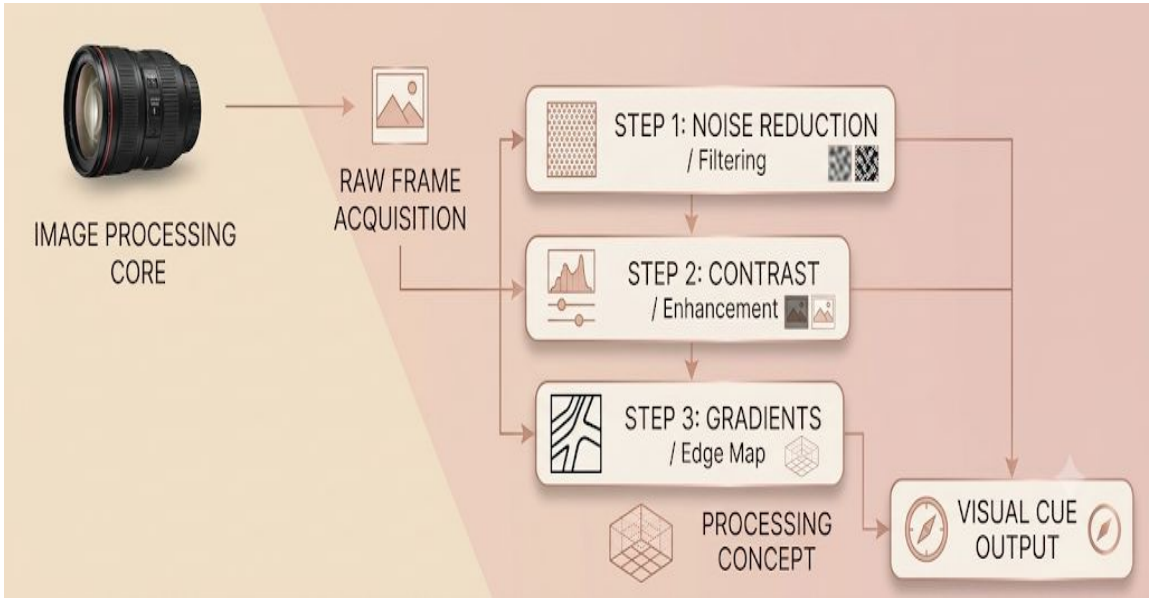
Improves brightness and contrast.

3. Filtering

Smooths or sharpens image structures.

4. Edge Detection

Identifies boundaries and structural changes.



Output

A cleaner image containing important visual structures for subsequent analysis.

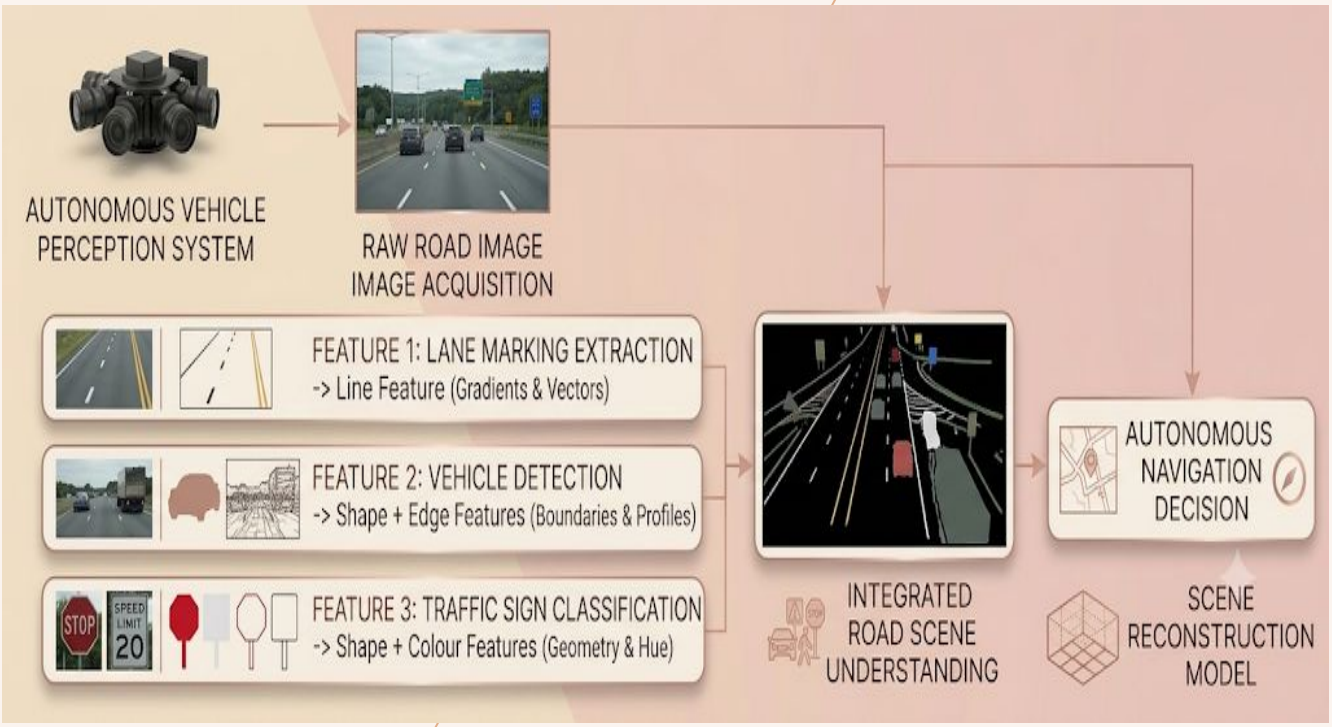
Feature Extraction :

From Pixels to Visual Features

After preprocessing, significant image features are extracted.

Features Used

- Edges
- Lines
- Corners
- Shapes
- Textures
- Colour information
- Geometric patterns



Purpose

Reduce complex visual information into **discriminative features** that can be processed efficiently.

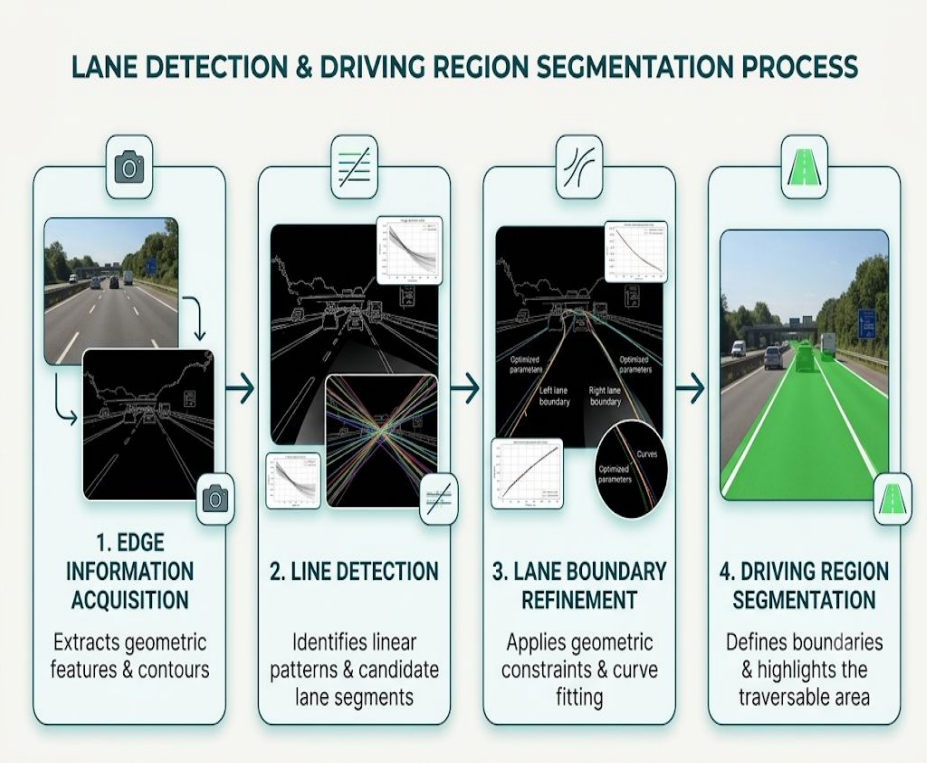
Mid-Level Vision :

Stage 3: Structural Interpretation

Mid-level vision transforms extracted features into meaningful **regions, structures, and spatial relationships**.

Major Operations:

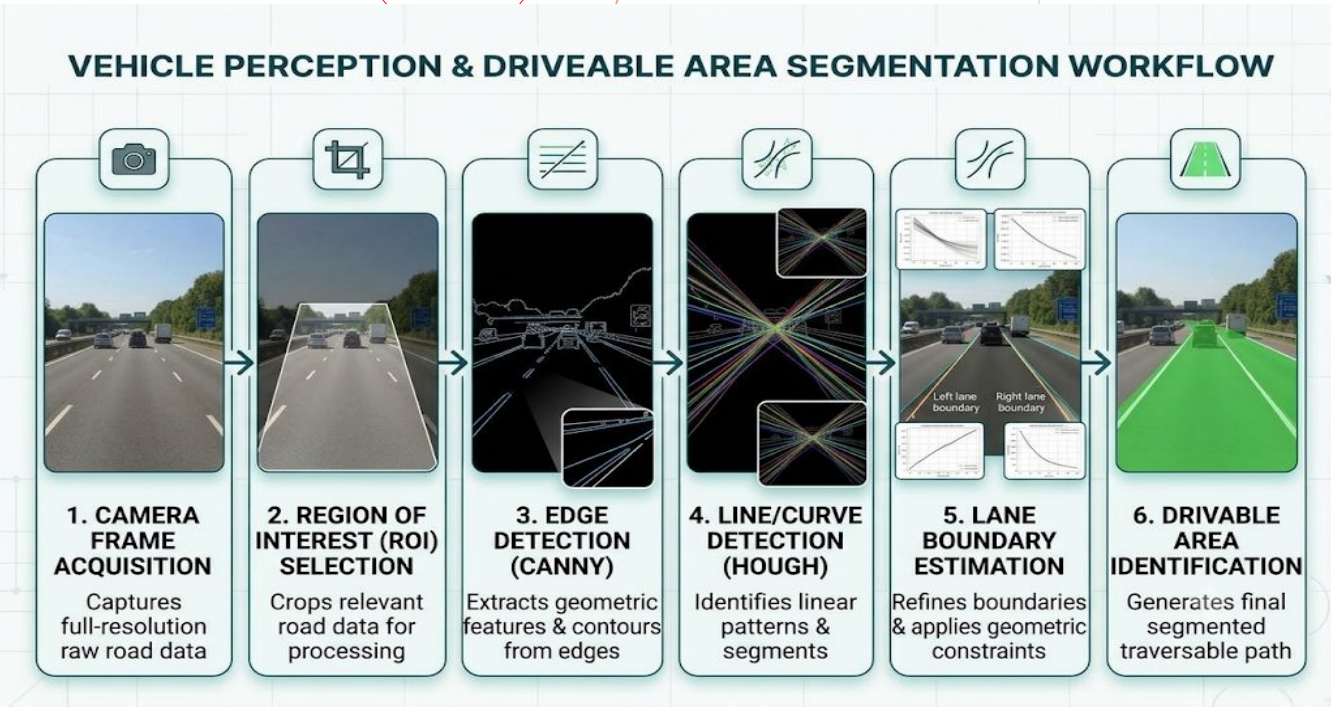
- Image segmentation
- Region extraction
- Lane detection
- Object localization
- Shape analysis
- Feature matching



Lane & Road Detection :

Understanding the Drivable Area

Lane detection is essential for determining the vehicle's position relative to the road.



System Output

- Current lane
- Lane boundaries
- Road direction
- Possible lane departure

Application

Supports **lane keeping and path planning**.

High-Level Vision :

Stage 4: Semantic Understanding

High-level Computer Vision interprets visual information in terms of **objects and their meaning**.

Major Tasks :

Task	Output
Object Detection	Location of objects
Classification	Object category
Recognition	Object identity/type
Semantic Segmentation	Pixel-level scene labels
Traffic Sign Recognition	Sign meaning

Example

The system converts:

Pixel region → “Car”

Pixel region → “Pedestrian”

Pixel region → “Traffic Signal”

Object Detection & Tracking :

Dynamic Object Perception

Autonomous vehicles must understand not only **what objects exist**, but also how they move.

Detection

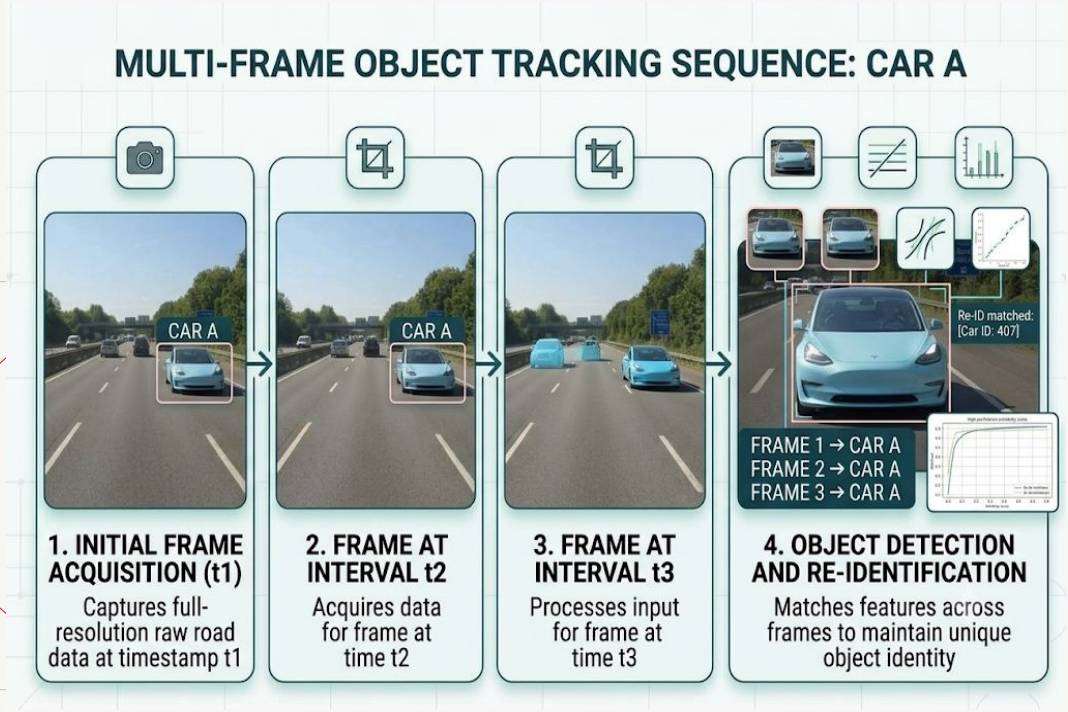
Identifies objects in each frame.

Input → Image

Output → Bounding Boxes + Class Labels

Tracking

Associates detected objects across consecutive frames.



Scene Understanding :

From Individual Objects to a Complete Scene

The vehicle combines information from multiple Computer Vision modules.

Example Road Scene

Detected elements:

- 🛣️ Lane boundaries
- 🚗 Vehicle ahead
- 🚶 Pedestrian
- 🚦 Traffic signal
- 🚧 Road obstacle

Scene Interpretation

Vehicle ahead is occupying the current lane, while a pedestrian is approaching the crossing area.

This represents a transition from object recognition to contextual understanding.

Navigation Decision :

From Perception to Driving Action

Visual information is used to support navigation and control.

Decision Pipeline

Object Detection

+

Lane Detection

+

Traffic Signal Recognition

+

Scene Understanding

↓

Navigation Decision

↓

Vehicle Control

Possible Actions

- Maintain speed
- Brake
- Accelerate
- Maintain lane
- Change lane
- Stop
- Avoid obstacle

Example

Pedestrian detected in vehicle path

→ Risk assessment

→ Reduce speed

→ Maintain safe distance

Case Study: Autonomous Vehicle at a Traffic Signal :

- The camera captures the road scene.
- **Low-level vision** removes noise and detects edges.
- **Mid-level vision** identifies lanes and the road area.
- **High-level vision** detects cars, pedestrians, and traffic signals.
- The system understands the traffic situation.
- **Red signal + pedestrian ahead → Vehicle stops safely.**
- **Final Result:** Camera Image → Vision Processing → Scene Understanding → Safe Action.

Conclusion :

- Computer Vision enables autonomous vehicles to **understand their surroundings** using camera images.
- **Low-level vision** improves images and extracts basic features.
- **Mid-level vision** identifies important structures such as lanes and road regions.
- **High-level vision** detects and recognizes vehicles, pedestrians, and traffic signs.
- These stages convert **raw images into meaningful information**.
- The information helps the vehicle make **safe and intelligent navigation decisions**.
- Thus, Computer Vision plays a **key role in autonomous driving and road safety**.

References

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